

Total Points

80.5 / 105 pts

Question 1

Q1

8 / 10 pts

– 0 pts Correct

✓ – 2 pts Partially correct

– 10 pts Incorrect or missed question

– 8 pts Unclear or incomplete answer

– 5 pts Incorrect or incomplete explanation

Question 2

Q2

10 / 10 pts

✓ – 0 pts Correct

– 10 pts Incorrect

– 5 pts Partially correct

– 2 pts Minor Mistakes

Question 3

Q3

10 / 10 pts

✓ – 0 pts Correct

– 10 pts Incorrect

– 5 pts Wrong definition

– 2 pts 2 steps right

– 4 pts 1 step right

– 5 pts Incorrect steps

– 2 pts Partially correct definiton

Question 4

Q4

7.5 / 15 pts

– 0 pts Correct

✓ – 7.5 pts half correct

– 15 pts Incorrect or too vague

Question 5

Q5

15 / 15 pts

✓ – 0 pts Correct

– 4 pts Minor Error

– 5 pts Mild Error

– 10 pts Significant Error

– 15 pts Incorrect

Question 6

Q6

0 / 15 pts

– 0 pts Correct

✓ – 15 pts Incorrect or too vague

– 5 pts minor error

Question 7

Q7

15 / 15 pts

✓ – 0 pts Correct

– 15 pts Incorrect/Blank

– 5 pts Partial

– 10 pts Minor Miss

Question 8

Q8

15 / 15 pts

✓ – 0 pts Correct

– 15 pts Incorrect

– 5 pts wrong total output

1. (10 points) What is *memory stall*?

A memory stall is a bad situation where the CPU is idly waiting for an I/O request from the running process. This is bad b/c the wasted machine cycles could be used on something else.

2. (10 points) What happens to the CPU when a running process requests to read a file? Why does it happen?

The process transitions to the waiting state and is put into the I/O queue, where it will remain until the request is complete. In the meantime, a process is popped from ready queue & is transitioned to running state and the CPU starts executing it. This happens to prevent memory stall - there is no point in the CPU idly waiting for the read request, when it can start executing another program.

3. (10 points) What is *machine cycle*? What are the steps of machine cycle (just name them, no need to describe them)?

The machine cycle is the series of steps the CPU takes to execute one instruction. The steps are the following: Fetch, decode, execute, and optionally, write back.

4. (15 points) What is *multitasking*? How is it arranged?

Multitasking is a technique the operating system uses to simulate simultaneous execution of multiple processes via context switching.

5. (15 points) What is API of the operating system?

The API of the operating system is the set of tools accessible to the programmer / user programs in order to interact w/ the operating system.

6. (15 points) What is logical disk?

A logical disk is a simulated partition of a hard disk. A hard drive can be split into multiple logical disks that each have their own memory space and even operating system.

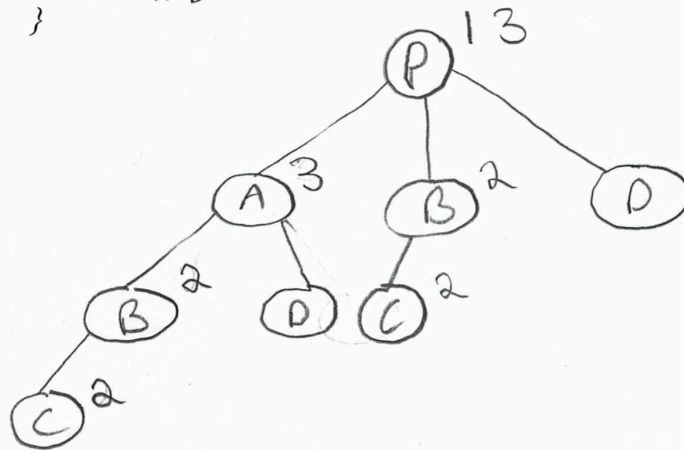
7. (15 points) What is *POSIX*?

POSIX is a set of standards that ensures compatibility between all UNIX-based operating systems. That way, programs that run and use API calls on one UNIX-based operating system can work on another UNIX-based operating system.

8. (15 points) Show process tree and any possible output of the following fragment of code

```
cout << '1' << endl;
fork(); // A

if (!fork()) { // B
    fork(); // C
    cout << '2' << endl;
}
else {
    cout << '3' << endl;
    fork(); // D
}
```



1
2
2
3
2
2
3